

9. (4 marks) Consider the following MIPS assembly code (where R0, R1, and R2 are 32-bit general purpose registers, and R0 has initialized with a value of zero):

```
i)   ADDI R0, R0, 7
ii)  LW   R1, 0x0003(R0)
iii) ADDI R0, R0, 5
iv)  SW   R2, 0x0002(R0)
```

Based on the code sequence, answer the following: (a). (1 mark) Show the 32-bit values (hexadecimal) of the register R0 after the executions of the instructions (i) and (iii). **0x0000000C** (b). (3 marks) Show the physical memory addresses (hexadecimal) of the instructions (ii) and (iv) **0x0000000A and 0x0000000E**.

10. (2 marks) Assume that a high-level language compiler is running on an MIPS system and it generates the following assembly code (where R0, R1, R2, R3, and R4 are 32-bit general purpose registers and are loaded with integer variables *f*, *g*, *h*, *i* and *j*, respectively). Show its source code. Find your answer

```
BNE R3, R4, Next
ADD R0, R1, R2
J Exit
Next:
SUB R0, R0, R3
Exit:
```

11. (3 marks) Based on the given instruction sequence, determine the 32-bit values (in hexadecimal) of the registers R1, R2, and R3 after the instruction execution (assume that the register R0 is initialized with zero). **R1 = 0x0000000A, R2 = 0x0000000A, and R3 = 0x0000000C**

```
1. ADDI R1, R0, 0x0005
2. SLL  R1, R1, 1
3. ADDI R2, R0, 0x000A
4. BEQ  R1, R2, NEXT
5. SUBI  R1, R0, 8
6. ADD   R3, R2, R0
7. NEXT: ADDI R3, R2, 0x0002
```

[21 marks] System Pipelining

12. (5 marks) What is **pipelining**? Briefly describe various pipeline stages and the functions of the MIPS system. Find your answer

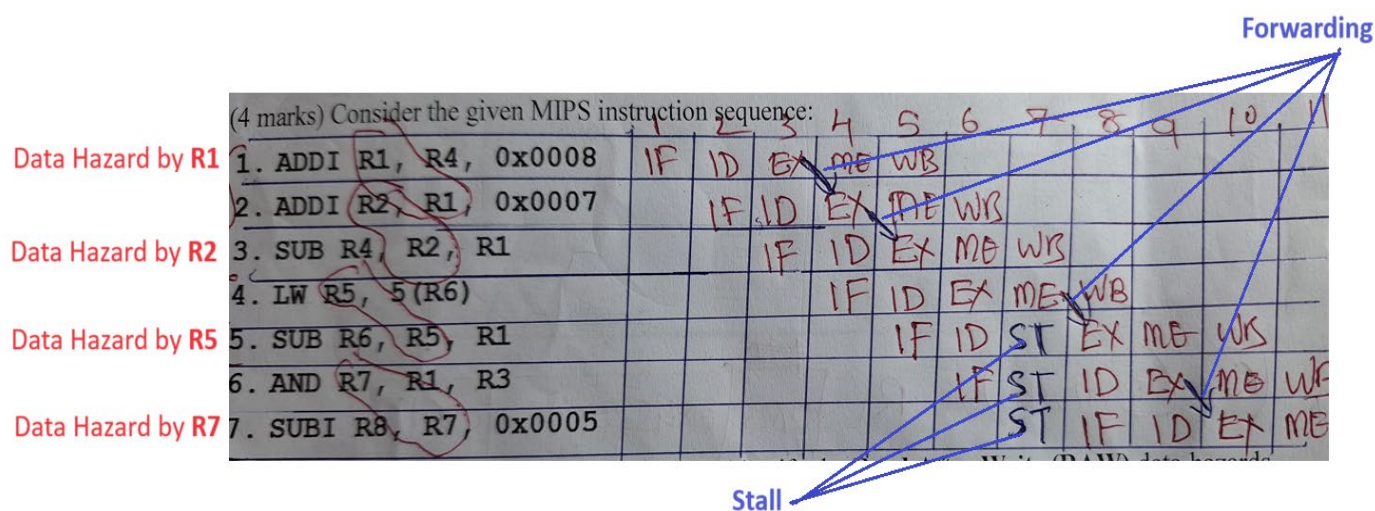
13. (2 marks) Consider an **S-stage** pipeline system executing a program with **K** instructions. Show that the **pipeline's speedup equals K** when **S** reaches infinity. Find your answer

14. (4 marks) What is a *structural hazard*? Consider a pipeline sequence diagram of the MIPS system, shown below. Is the stall (marked by a circle) in clock cycle four by **instruction *i*+3** due to a structural hazard? If your answer is yes, describe one of the possible ways to eliminate the stall from the pipeline. **Resource conflicts in a pipeline execution are called structural hazards. Yes, the stall is the result of the structural hazard. The solution is to use different memory for IF and MEM stages or rearrange the non-dependent instruction to avoid stalls due to structural hazards.**

Clock Cycle number									
Instruction	1	2	3	4	5	6	7	8	9
Load instruction	IF	ID	EX	MEM	WB				
Instruction <i>i</i> + 1		IF	ID	EX	MEM	WB			
Instruction <i>i</i> + 2			IF	ID	EX	MEM	WB		
Instruction <i>i</i> + 3				stall	IF	ID	EX	MEM	WB

15. (4 marks) Consider the given MIPS instruction sequence:

1. ADDI R1, R4, 0x0008
2. ADDI R2, R1, 0x0007
3. SUB R4, R2, R1
4. LW R5, 5(R6)
5. SUB R6, R5, R1
6. AND R7, R1, R3
7. SUBI R8, R7, 0x0005



Using the MIPS’s **pipeline sequence diagram**, identify the **Read After Write (RAW)** data hazards among the instructions (if any) by indicating the **hazard-causing registers with their instruction numbers**. If any RAW data hazard is detected among instructions, show the various solution steps taken by the MIPS system (where R1, R2, R3, R4, R5, R6, R7, and R8 are 32-bit general purpose registers).

16. (4 marks) Consider an **unpipelined** processor. It has a **2 ns** clock cycle and uses **four** cycles for ALU operations, **five** for branches, and six for memory operations. Assume that the relative frequencies of these operations are 50%, 15%, and 35%, respectively. **Pipelining** the processor adds **0.4 ns** of overhead to the clock (ignore other pipeline latencies and hazards). Find out how much speedup the instruction execution rate will gain from the pipeline.

Avg.Inst.Exe.time(un-pipeline) = 9.7ns, Avg.Inst.Exe.time(pipeline) = 2.4ns.

Speedup = 9.7ns/2.4ns = 4.042 times

17. (2 marks) Assume that the MIPS pipelined system has imposed a stall of **0.78** clock cycles per instruction for a particular user process. What is the **system's speedup**? **5/(1+0.78) = 2.809**

[10 marks] **Memory System**

18. (2 marks) Why does the MIPS system use a **5-bit address** to access its register file? **Because of its 32 registers**
19. (4 marks) Consider the following MIPS instruction (where R0 and R1 are two 32-bit general-purpose integer registers): **SW R1, 0x0010(R0)** . Assume that the **R0** and **R1** are initialized with **two** and **0x56A78BC1**, respectively. After the **SW** operation, show the values of the **MIPS’s byte-addressable data memory** (a section of the **data memory** is shown below).

Address	Word
0x00000010	?
0x00000011	?
0x00000012	0x56
0x00000013	0xA7
0x00000014	0x8B
0x00000015	0xC1

- 20.** (4 marks) Below is a section of the MIPS pipeline system's **instruction memory (IM)**. Based on that:
- (a). (1 mark) What is the **Program Counter (PC)** value if the **IF stage** selects the instruction **0x3B1ED031**? **0x00000004**
 - (b). (1 mark) Show the instruction selected in the **previous clock cycle**. **0x8AF36752**
 - (c). (1 mark) Give the value of its **PC value** (from (b)). **0x00000000**
 - (d). (1 mark) Why is the PC important in a processor? **PC is an instruction pointer register, and without the PC, the CPU is unable to track and fetch instructions from the Instruction Memory (IM).**

Memory Address	Memory Word
0x00000000	0x8A
0x00000001	0xF3
0x00000002	0x67
0x00000003	0x52
0x00000004	0x3B
0x00000005	0x1E
0x00000006	0xD0
0x00000007	0x31
0x00000008	0x41
0x00000009	0xFA
0x0000000A	0x8C
0x0000000B	0x1C

===== End =====